

Carbon Cycle Simulation and Climate Scenarios

Our house is burning down and we're blind to it

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1. Problem Formulation

The Cauchy Problem :

We model the global carbon cycle as a dynamical system with a state vector $Y(t) \in \mathbb{R}^8$ representing the 8 carbon reservoirs:

$$Y(t) = [\text{Atmosphere}, \text{CarbonateRock}, \text{DeepOcean}, \text{FossilFuel}, \\ \text{Plants}, \text{Soils}, \text{SurfaceOcean}, \text{VegLandArea}]^T$$

System governed by an ODE:

$$\begin{cases} \frac{dY}{dt} = f(Y, t) \\ Y(t_0) = Y_0 \quad (\text{Pre-industrial conditions at } t_0 = 1850) \end{cases}$$

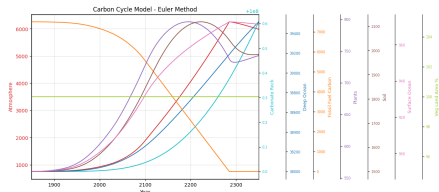
Project Objective :

- Solve this Cauchy problem numerically.
- Simulate and project the GlobalTemp variable from 1850 up to the year 2300.

2. Numerical Methods: Euler vs RK4

Explicit Euler Order 1 ($\mathcal{O}(dt)$)

$$Y_{n+1} = Y_n + dt \cdot f(Y_n, t_n)$$



Runge-Kutta 4 Order 4 ($\mathcal{O}(dt^4)$)

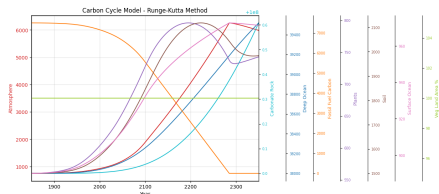
$$k_1 = f(Y_n, t_n)$$

$$k_2 = f(Y_n + \frac{dt}{2} k_1, t_n + \frac{dt}{2})$$

$$k_3 = f(Y_n + \frac{dt}{2} k_2, t_n + \frac{dt}{2})$$

$$k_4 = f(Y_n + dt \cdot k_3, t_n + dt)$$

$$Y_{n+1} = Y_n + \frac{dt}{6} (k_1 + 2k_2 + 2k_3 + k_4)$$



3. Convergence Proof: Euler & RK4

Lipschitz Continuity:

Equations are C^∞ . Thus, f is L -Lipschitz. Cauchy-Lipschitz theorem : unique exact solution $Y^*(t)$.

Explicit Euler

Consistency:

$$\epsilon_n = \frac{dt}{2} Y^{*'''}(t_n) + \mathcal{O}(dt^2) = \mathcal{O}(dt).$$

(Order 1)

Stability:

$$\|Y_{n+1} - \tilde{Y}_{n+1}\| \leq (1 + dt \cdot L) \|Y_n - \tilde{Y}_n\|$$

Iterating n times yields bounded growth $\leq e^{LT}$.

Runge-Kutta 4

Consistency: $\epsilon_n = \mathcal{O}(dt^4)$. (Order 4)

Stability: The intermediate steps k_i inherit the Lipschitz property of f . Similar bounded exponential growth guarantees stability.

Convergence (Lax Equivalence Theorem): Consistency + Stability $\implies$ Convergence.

4. Empirical Validation

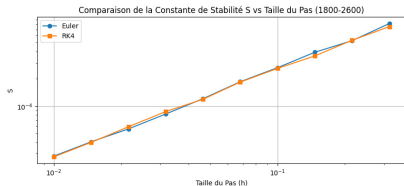


Fig 1: Euler error vs dt (Slope ≈ 1)



Fig 2: RK4 error vs dt (Slope ≈ 4)

Stability :

- We plotted the stability constant S as a function of the step size h for both methods. We observe that S is quite similar for both and increases as the step size increases.

Consistency :

- We plotted the sum of the relative consistency errors for different time steps h using the formula:

$$E_{cons}(h) = \sum_{n=0}^{N-1} \epsilon_n(h)$$

- As expected, as the step size h decreases, the error tends towards 0, empirically proving the consistency

5. Conclusion

Validation :

- RK4 robust ($\mathcal{O}(dt^4)$).
- Reliable physical model.

Temperature :

- The temperature was modeled based on certain accepted hypotheses.

Conclusion : Urgency confirmed (Bibi is right !).

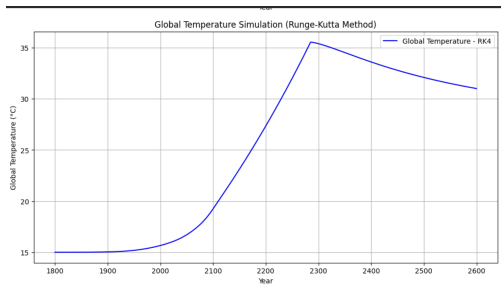


Fig 1. Évolution de la température (1850-2300)